



Contact



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The Hong Kong University of Science
and Technology
Clear Water Bay, Kowloon
Hong Kong

Education

Ph.D. in Civil Engineering
(Expected Aug 2029)

The Hong Kong University of Science
and Technology

M.E. in Environmental Engineering
(Jun 2024)

University of Science and Technology
Beijing, China; University of Birmingham,
United Kingdom
GPA: 3.9/4.0

B.E. in Environmental Engineering
(Jun 2021)

University of Science and Technology
Beijing, China
GPA: 3.4/4.0

Skills & Expertise

- Anaerobic digestion experimental design
- Feature engineering for biological processes
- Machine learning model development
- Analytical chemistry
- Biowaste experimental analysis and characterization

Curriculum Vitae

Jie Meng

Graduate student in Civil and Environmental Engineering, dedicated to sustainable waste management, interest in integrating hands-on anaerobic digestion (AD) experimentation with machine learning prediction to enhance biowaste valorization efficiency. Focus on bridging biological treatment processes and data-driven modeling for next generation resource recovery systems.

Professional Experience

Sep 2026–Current

Graduate Research and Teaching Assistant

Department of Civil and Environmental Engineering
The Hong Kong University of Science and Technology,
Hong Kong

Research focus: Machine learning integration in AD
system to enhance waste valorization efficiency.

PI: Prof. Samir Kumar Khanal

Sep 2024–Aug 2025

Research Assistant

The Hong Kong Polytechnic University, Hong Kong
Research focus: Photothermal catalytic system for
efficient separation and recycling of waste blended
fabrics

Supervisor: Prof. Jintu Fan, Dr. Hanchao Zhang

Sep 2021–Jun 2024

Graduate Research Assistant

University of Science and Technology Beijing, China
University of Birmingham, United Kingdom

Research focus: Solid waste resource and energy
utilization, low temperature fuel cell and electrolytic
cell

Supervisor: Prof. Qunhui Wang & Prof. Shangfeng Du

Research Areas

- Machine learning for anaerobic digestion modeling
- Biowaste valorization for bioenergy recovery
- Data-driven process monitoring and fault prediction
- Integrated sustainability assessment of waste treatment systems



Publications

1. Hanchao, Z., Qiang, Y., Yudong, W., **Jie, M.**, Xiaoyang, G., Yi, J., ... & Jintu, F. (2025). Bilayer Solar Crystallizer by a Directional Liquid Transport Fabric for Stable Brine Treatment and Ion Recycling. *Advanced Functional Materials*, 35(52), e11412.
2. Xu, M., Yang, M., Sun, H., **Meng, J.**, Li, Y., Gao, M., ... & Wu, C. (2022). Role of multistage inoculation on the co-composting of food waste and biogas residue. *Bioresource Technology*, 361, 127681.
3. Xie, D., Yang, M., Zhang, S., Xu, M., **Meng, J.**, Wu, C., ... & Liu, S. (2023). In-situ utilization of nitrogen-rich wastewater discharged from a biotrickling filter as a moisture conditioning agent for composting: Insights into nitrogen transformation behavior and microbial mechanism. *Bioresource Technology*, 369, 128362.
4. Xu, M., Sun, H., Yang, M., Xie, D., Sun, X., **Meng, J.**, ... & Wu, C. (2022). Biodrying of biogas residue through a thermophilic bacterial agent inoculation: insights into dewatering contribution and microbial mechanism. *Bioresource technology*, 355, 127256.
5. **MENG, J.**, ZHU, X., XU, M., RONG, L., WU, C., & WANG, Q. (2025). Experimental Study and Modeling Analysis on the Removal of Simulated Ammonia Containing Odor by Decomposed Cow Manure Residue. *Ecology and Environmental Sciences*, 34(6), 922. (in Chinese)

Awards and Honors

1. Scholarship of China Scholarship Council (CSC)- Joint Training Master Program-University of Birmingham - University of Science and Technology Beijing – (Nov 2022–May 2023)
2. The First prize (First author) - The 13th National College Students' Energy Conservation and Emission Reduction Competition
3. The First prize (First author) - The 17th "Challenge Cup" National extracurricular academic and technological works competition
4. The Grand prize (First author) - The 11th "Challenge Cup" extracurricular academic and technological works competition Beijing
5. The Grand prize (First author) - The 2nd Beijing College Students' Social Practice and Science and Technology Competition on energy conservation, water conservation, and low carbon emission reduction
6. First Honor Academic Scholarship – University of Science and Technology Beijing – (Sep 2021–Jun 2023)
7. Outstanding student cadre – University of Science and Technology Beijing – (2022)
8. The Gold Award - The "Academic Three Minute" Speech Competition - University of Science and Technology Beijing – (2022)
9. Technology Star - University of Science and Technology Beijing – (2022)
10. Outstanding graduate student – University of Science and Technology Beijing – (2021)
11. Outstanding Student Athlete – University of Science and Technology Beijing – (2020)
12. Sponsorship of Procter&Gamble China Pioneer Program – (2019)



Certifications

1. Responsible Conduct of Research (1-RCR), The Hong Kong University of Science and Technology, Jan 2026
2. Lab Safety Training on Chemical, Biological and Fire Safety, Nov 2025

Relevant Courses

Postgraduate Courses

(M.E. in Environmental Engineering)

- Environmental Biochemistry and Molecular Biology
- Principles and Technologies of Wastewater Treatment
- Biomass Science and Engineering
- Adsorption Separation
- Frontiers in Soil Biology
- Mathematical Modeling

Postgraduate Courses

(Ph.D. in Civil and Environmental Engineering)

- Practical Machine Learning
- Process Design of Environmental Engineering Facilities
- Applied AI for Sustainability Research
- Aquatic Chemistry
- Resource Recovery from Bioresources

Undergraduate Courses

(B.E. in Environmental Engineering)

- Principles of Environmental Engineering
- Solid Waste Treatment And Disposal
- Water Pollution Control Engineering
- Air Pollution Control Engineering
- Physical Pollution Control
- Inorganic/organic/Physical/Analytical Chemistry
- Microbiology for Environmental Engineering
- Computer Aided Design for Environmental Engineering
- Environmental Monitoring
- Cleaner Production and Circular Economy
- Environmental Impact Assessment Practice
- Environmental Toxicology